

Questions

Q1.) Simplify the following surds:

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|-----------------|------------------|------------------|
| (a) $\sqrt{24}$ | (b) $\sqrt{80}$ | (c) $\sqrt{75}$ |
| (d) $\sqrt{52}$ | (e) $\sqrt{162}$ | (f) $\sqrt{180}$ |

Q2.) (a) Write 1008 as a product of its prime factors.

(b) Use part (a) to simplify the surd $\sqrt{1008}$.

Q3.) Simplify the following expressions:

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|----------------------------|----------------------------|---|
| (a) $3\sqrt{2} + \sqrt{2}$ | (b) $\sqrt{5} + \sqrt{20}$ | (c) $4\sqrt{12} + 2\sqrt{6} - 5\sqrt{75}$ |
|----------------------------|----------------------------|---|

Q4.) Evaluate the following in their simplest forms:

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|----------------------------------|-----------------------------------|-----------------------------------|------------------------------------|
| (a) $\sqrt{7} \times \sqrt{7}$ | (b) $(\sqrt{6})^2$ | (c) $(\sqrt{11})^2$ | (d) $(\sqrt{1275})^2$ |
| (e) $\sqrt{2} \times \sqrt{3}$ | (f) $\sqrt{5} \times \sqrt{10}$ | (g) $\sqrt{35} \times \sqrt{14}$ | (h) $2\sqrt{3} \times \sqrt{3}$ |
| (i) $5\sqrt{5} \times 6\sqrt{5}$ | (j) $\sqrt{11} \times 4\sqrt{22}$ | (k) $3\sqrt{6} \times 2\sqrt{15}$ | (l) $-3\sqrt{2} \times -5\sqrt{3}$ |

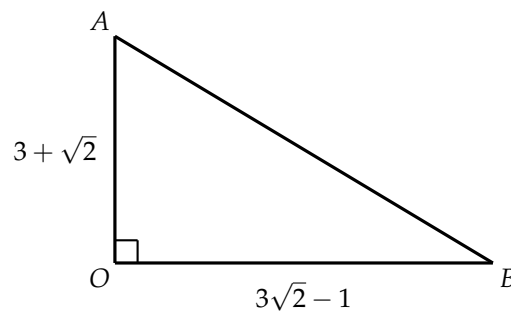
Q5.) Expand and simplify:

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|------------------------------|--------------------------------------|--------------------------------|
| (a) $\sqrt{2}(\sqrt{2} + 3)$ | (b) $\sqrt{5}(\sqrt{10} + \sqrt{5})$ | (c) $3\sqrt{3}(1 + 5\sqrt{3})$ |
|------------------------------|--------------------------------------|--------------------------------|

Q6.) Expand and simplify:

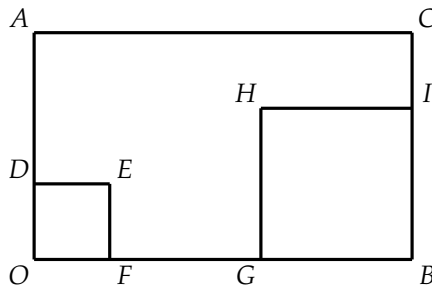
- | | |
|--|--|
| (a) $(1 + \sqrt{2})(1 + \sqrt{3})$ | (b) $(2 + \sqrt{5})(3 + \sqrt{5})$ |
| (c) $(2 + \sqrt{5})(3 - \sqrt{5})$ | (d) $(3 + 2\sqrt{7})(1 + 4\sqrt{7})$ |
| (e) $(8 + \sqrt{11})^2$ | (f) $(5 - 2\sqrt{5})^2$ |
| (g) $(\sqrt{7} + 1)(\sqrt{7} - 1)$ | (h) $(\sqrt{6} + 2)(\sqrt{6} - 2)$ |
| (i) $(\sqrt{3} + \sqrt{2})(\sqrt{3} - \sqrt{2})$ | (j) $(2\sqrt{10} + 4)(2\sqrt{10} - 4)$ |

Q7.) Calculate the **exact** length of side AB for the triangle below:



Q8.) In the diagram below,

- The area of square $ODEF$ is 2cm^2 .
- The area of square $GHIB$ is 8cm^2
- The length of line segment FG is 4cm .
- The length of line segment CI is 2cm .



Calculate the **exact** area of rectangle $OACB$.

Q9.) Rationalise the denominators of the following fractions:

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|---------------------------------|-----------------------------------|------------------------------------|
| (a) $\frac{1}{\sqrt{10}}$ | (b) $\frac{5}{2\sqrt{3}}$ | (c) $\frac{2}{3\sqrt{7}}$ |
| (d) $\frac{\sqrt{3}}{\sqrt{5}}$ | (e) $\frac{1+\sqrt{2}}{\sqrt{6}}$ | (f) $\frac{2+\sqrt{3}}{7\sqrt{3}}$ |

Q10.) Rationalise the denominators of the following fractions:

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|-------------------------------------|--------------------------------------|--------------------------------------|
| (a) $\frac{1}{2+\sqrt{3}}$ | (b) $\frac{5}{6-\sqrt{2}}$ | (c) $\frac{\sqrt{5}}{5+\sqrt{5}}$ |
| (d) $\frac{2\sqrt{3}}{10-\sqrt{3}}$ | (e) $\frac{1+\sqrt{2}}{1-2\sqrt{2}}$ | (f) $\frac{3+4\sqrt{7}}{\sqrt{7}-2}$ |

Q11.) Rationalise the denominator of the fraction

$$\frac{3+\sqrt{2}}{\sqrt{3}+\sqrt{2}}$$

Q12.) **Challenge:** Given that $\sqrt{4\sqrt{2}+6}$ can be written in the form $a+b\sqrt{2}$, where a and b are whole numbers, find the values of a and b .

Hint: Since $a+b\sqrt{2} = \sqrt{4\sqrt{2}+6}$, try squaring both sides and matching up the coefficients.